

Riphah International Colleges

A Project of Riphah International University

Course Outline (Latest Updated)

Course Information	Course Title		Computer Organization and Assembly Language (COAL)				
	Course ID		Course Type		Core		
	Credit hours		3 (2-3)		Hours per week (C-L) 5 (2-3)		
	Program(s)		ADP (Computing) Specialization in Computer Sciences		Preferred Semester 3 rd		
	Date		Implemented from 29-Oct-24		Version V 2.0		
Course Description	- The main objective of this course is to introduce the organization of computer systems and usage of assembly language for optimization and control. Fundamentals of computer design; quantifying cost and performance; instruction set architecture; program behavior and measurement of instruction set use; processor data paths and control; pipelining, handling pipeline hazards; memory hierarchies and performance; I/O devices, controllers and drivers; I/O and system performance.						
Course Objectives (CO)	The objective of this course is to enable students to understand;						
	No.	Objective				Relation with PEO	
	CO1.	Basics of computer architecture					
	CO2.	Impact on compiler & OS design					
	CO3.	Introduction to Computer Systems Design					
	CO4	How to build fast systems					
CO5.	Computer performance measurement						
Course Learning Outcomes (CLO)	At the end of this course students will be able to;						
	No.	Outcome			Relation with PLO	BT Level	PLO Level
	CLO1	Identify the function units of computer architecture, different computer architecture			PLO1		
	CLO2	Differentiate the various instructions types, different secondary storage devices and addressing modes used for programming			PLO2		
	CLO3	Infer the interrupts and direct memory access the standard I/O interfaces.			PLO2		
	CLO4	Illustrate the design of a pipelined CPU and cache hierarchy, the need for data path & control.			PLO3		
	CLO5	Interpret between RAM, ROM, speed, cost, size, cache & virtual memory and need for improving cache performance.			PLO5		
Lecture type	Class room Lectures, Lab Sessions, Project Presentation						
Prerequisites	Digital Logic and Design						
Follow up Courses							
Textbook	Title		Edition	Authors	Publisher	Year	ISBN
	Computer Organization and Architecture		10th Edition	William Stallings	Prentice Hall Inc.		
Reference Books	Computer System Architecture		Latest Edition	M. Morris Mano			
	Computer Systems: A Programmer's Perspective		3/E (CS:APP 3e)	Randal E. Bryant and David R.O' Hallaron,	Carnegie Mellon University		

	<u>Computer System Architecture</u>	3rd Edition	Morris Mano	Prentice Hall		
	<u>Assembly Language Programming for Intel- Computer</u>	Latest Edition				
	<u>Structured Computer Organization</u>		Morris Mano	Prentice Hall Inc.		
Reference Material	- Link for William Stalling Resources : http://williamstallings.com/ComputerOrganization/styled-7/ http://williamstallings.com/ComputerOrganization/styled-6/ (Student/Teacher resources) - Other supporting material web links: https://easyengineering.net/computer-system-architecture-by-mano-m-morris-nw/91zknsr8cbl/ https://github.com/coderplex-org/learn/blob/master/computer-science/Learn-CS.md - Links are provided on Books names.					
Course Software or Tool	Assembly Language Tools required for Lab https://spimsimulator.sourceforge.net/ (MIPS32 Simulator Link) https://rivoire.cs.sonoma.edu/cs351/wemips/					

GRADE DISTRIBUTION:

Evaluation Type	Percentage %	Activities
Assignments and Presentations	10 %	Min 4 in the semester
Quiz & Projects	10 %	Min 4 in the semester
Lab	15 %	On weekly basis
Mid-Term	25 %	Contents from Week-1 to Week-8
Final-Term	40 %	Contents from Week-10-Week-17
Total Points	100	
Methods of Evaluation	Quizzes, Assignments, Mid and Final Term, Lab Paper, Lab-Projects	
Note:	Labs are managed and evaluated separately	

General Classroom Norms:

- ♦ Class attendance is mandatory. You may miss up to 25% (8 out of 34 sessions) class sessions but save it for emergency only.
- ♦ In case you exceed this level, you will be withdrawn from the course.
- ♦ As a courtesy to the instructor and other students, be prepared to arrive at class and be in your seat on time.
- ♦ In addition, please note that each class lasts for 90 minutes (1.5 Hours).
- ♦ Cell phones should be powered off or kept on silent mode.
- ♦ Eatables should be avoided in the class.
- ♦ Disruptive behaviors are not acceptable in the class.
- ♦ The Dress Code defined by Campus Administration has to be observed, no warnings will be given, and violators will be referred to the discipline committee for further actions.
- ♦ Any other rules defined by campus are also mandatory to be followed.

COURSE OUTLINE

(COMPUTER ORGANIZATION AND ASSEMBLY LANGUAGE)

Week No.	Topic	Lecture No.	Relation with CLO	Lecture Material	Class Activity	Tasks
W1.	Organization and Architecture, Why Study Computer Organization and Arch, Structure and Function	L1.	CLO1	Slides: Chapter 0 Slides: Chapter 1	Activity: Introduction of each student	Homework: Go through the module handbook thoroughly
		L2.	CLO1	Slides Lecture 1		
W2.	Structure and Function, Computer Evolution and performance, The First Generation: Vacuum Tubes	L3.	CLO1	Slides: Chapter 1		
		L4.	CLO1, CLO2			Assignment 1:
W3.	Von Neumann Structure, Expanded Structure of IAS Computer	L5.	CLO2, CLO3	Slides: Chapter 1		
		L6.	CLO2		Quiz 1	
W4.	Second Generation, Later Generation, Atop-Level view of Computer Function and Interconnection, Computer Functions	L7.	CLO4	Slides: Chapter 2		
		L8.	CLO3			
W5.	Computer Function, Instructions Fetch and Execute, Interrupts	L9.	CLO4, CLO5	Slides: Chapter 3		
		L10.	CLO5			
W6.	I/O Functions, Interconnection Structures, Bus Interconnection	L11.	CLO3	Slides: Chapter 3		Assignment 2:
		L12.	CLO3		Quiz 2	
W7.	BUS Structure, Multiple Bus Hierarchies, Elements of Bus Design,	L13.	CLO4	Slides: Chapter 3		
		L14.	CLO4			
W8.	MIPS Instructions, MIPS Addressing Modes,	L15.	CLO5	Slides: Chapter 15		
		L16.	CLO5			
W9.		L17.	Mid Term Exam Week			
		L18.				
W10.	Pipeline Implementation, MIPS Pipeline Stages, Hazards	L19.	CLO5	Slides: Chapter 15		
		L20.	CLO5			
W11.	Cache Memory, Computer Memory system Overview, Characteristics of Memory System, Memory Hierarchy	L21.	CLO5	Slides: Chapter 4		
		L22.	CLO5			Assignment 3:
W12.	Cache Memory Principles, Elements of Cache Design, Cache Addresses, Cache Size, Mapping Function	L23.	CLO5	Slides: Chapter 4	Quiz 3	
		L24.	CLO5			
W13.	External Memory, RAID, Input/ Output, External Devices, I/O Modules, I/O module structure	L25.	CLO5	Slides: Chapter 6		
		L26.	CLO3, CLO5			
W14.	Programmed I/O, Overview of Programmed I/O, I/O Commands, I/O Instructions,	L27.	CLO3	Slides: Chapter 7		

Week No.	Topic	Lecture No.	Relation with CLO	Lecture Material	Class Activity	Tasks
	Interrupt-Driven I/O, Interrupt Processing, Direct Memory Access	L28.	CLO3			
W15.	Drawbacks of Programmed & interrupt driven I/O, DMA Function, I/O Channels and Processors, The External Interface: Firewire and InfiniBand	L29.	CLO3	Slides: Chapter 7		Assignment 4:
		L30.	CLO3		Quiz 4	
W16.	Computer Arithmetic, The Arithmetic and logic unit, Integer Representation, Integer Arithmetic, CPU Structure and Function Processor Organization, Register Organization, Instruction Cycle	L31.	CLO4	Slides: Chapter 10		
		L32.	CLO3, CLO4			
W17.	Instruction Pipelining, Control Unit Operations, Micro Instructions, Control of the Processor	L33.	CLO5	Slides: Chapter 14		
		L34.	CLO5			
W18.		L35, L36.	Final Term Exam			

Computer Organization and Assembly Language LAB COURSE OUTLINE

S. NO.	WEEK NO.	TOPIC	ACTIVITIES/TASKS
1.	W1	Explore Visible Virtual Machine (VVM).	
2.	W2	Learn VVM Programming and simulate VVM program.	
3.	W3	VVM Programs using “BRP”, “BRZ” & “BR” instructions.	Assignment 1
4.	W4	Explore debugger using different commands	Quiz 1
5.	W5	Basic concept and functionality of Assembly Language 8086. (Basic Commands, Directories, Character Input and Display Service Codes)	
6.	W6	String display and enter instruction.	Assignment 2
7.	W7	Arithmetic operations in Assembly Language 8086.	Quiz 2
8.	W8	Shift Left and Shift Right Operations.	
9.	W9	MID TERM PAPERS	
10.	W10	Conditional and Unconditional Instructions in Assembly Language	
11.	W11	Implementation of If-Then-Else Instruction in Assembly Language	Assignment 3
12.	W12	Implementation of Loop Instruction in Assembly Language	Quiz 3
13.	W13	Concept of the STACK in Assembly Language	
14.	W14	Concept of the SWITCH Statement in Assembly Language	Assignment 4
15.	W15	GUI Implementation in Assembly Language	Quiz 4
16.	W16	PRESENTATIONS AND LAB ASSIGNMENTS	
17.	W17	LAB PAPERS	
18.	W18	FINAL TERM PAPERS	